Project 4: Contribute to OpenCV

The project will count as the equivalent of TWO projects in the final grading.

Artificial Intelligence tools can generate code quickly, but they cannot replace real engineering ability. In real software development, you still need to read large codebases, debug problems, test carefully, collaborate with others, and write reliable code.

For this project, instead of writing another isolated homework program, you will contribute to a real open-source project: **OpenCV**.

OpenCV is one of the most widely used computer vision libraries in the world. It is written mainly in C++, used in academia and industry, and maintained by developers worldwide.

Your goal is to study the OpenCV contribution workflow, modify its source code, and submit a meaningful Pull Request (PR) **after instructor review**.

Resources:

- Official guide of OpenCV: https://github.com/opencv/opencv/wiki/How_to_contribute
- You can find some issues to be fixed at: https://github.com/opencv/opencv/issues
- If you have no idea on the contribution, you can also choose RISC-V optimization for OpenCV functions. The description about RISC-V optimization can be found at https://github.com/opencv/opencv/wiki/GSoC_2026#idea-risc-v-optimizations

Requirements

1. Create a GitHub account if you do not have one.
2. Read the OpenCV contribution guide.
3. Fork the OpenCV repository to your own GitHub account.
4. Clone your fork locally.
5. Build OpenCV successfully on your computer.
6. Create a new branch for your work.
7. Complete **one** contribution task (Level A / B / C).
8. Submit your work to the instructor.

Step-by-Step Starter Guide (For Beginners)

If you have never used GitHub before, follow these steps.

Step 1: Fork OpenCV

Open website in your browser:

```
https://github.com/opencv/opencv
```

Click **Fork**.

This creates:

```
yourname/opencv
```

Step 2: Clone Your Fork

```
git clone https://github.com/yourname/opencv.git
cd opencv
```

Step 3: Add Official Repository

```
git remote add upstream https://github.com/opencv/opencv.git
```

Check:

```
git remote -v
```

Step 4: Create Your Own Branch

```
git checkout -b project4_yourname
```

Step 5: Build OpenCV

Use CMake + Visual Studio / Linux Makefile / macOS Xcode.

Step 6: Modify Files

Change only a small number of files.

Step 7: Commit

```
git add .
git commit -m "My test to improve error message in foo()"
```

Step 8: Push

```
git push origin project4_yourname
```

Contribution Levels

Choose **one** level.

Level A (Beginner Friendly)

Suitable for students with weak C++ background or no GitHub experience.

Your goal is: Successfully modify OpenCV and learn the workflow.

Recommended Tasks

A1. Fix Typos in Docs / Comments

Examples:

- grammar mistakes
- spelling mistakes
- unclear comments

Search files:

```
doc/
modules/*/*.cpp
modules/*/*.hpp
```

A2. Improve Error Messages

Example:

Old:

```
CV_Error(Error::StsBadArg, "Empty image");
```

Better:

```
CV_Error(Error::StsBadArg,
"Input image is empty. Please check imread() path.");
```

Search keyword:

```
CV_Error(
```

A3. Add a Small Sample Program

Example:

```
samples/cpp/
```

Write demo for:

- blur()
- resize()
- threshold()
- canny()

A4. Add One Simple Test

Search:

```
modules/*/test/
```

Example:

Test if resize() works on 1x1 image.

Minimum Requirement for Level A

1. Build OpenCV successfully.

2. Modify at least one file.

3. Commit successfully.
4. Report clearly what you changed.

Level B (Intermediate)

Suitable for students who understand C/C++ basics.

Your goal is:

> Solve a small real engineering problem.

Recommended Tasks

You can do something similar but not exactly the same as the following examples.

B1. Improve Parameter Checking

Examples:

- negative width / height
- null pointer
- invalid kernel size

Search:

```
CV_Assert(
```

B2. Fix Small Bugs

Examples:

- corner case crash
- wrong boundary behavior
- integer overflow risk
- warning in compiler

Try searching GitHub Issues at https://github.com/opencv/opencv/issues with keywords:

```
bug small fix
```

B3. Refactor Repeated Code

Find repeated code blocks and convert to a function.

B4. Small Performance Improvement

Examples:

- avoid repeated allocation
- move variable outside loop
- reduce copy operation

Minimum Requirement for Level B

1. Build OpenCV.

2. Modify code logic.

3. Test before / after.

4. Explain why your fix is useful.

Level C (Advanced)

Suitable for strong students.

Your goal is: Make a useful real contribution.

Recommended Tasks

C1. Add New Utility Function

C2. SIMD / Parallel Optimization

Use:

- Universal Intrinsics designed by OpenCV
- SSE / AVX
- ARM NEON
- OpenMP / TBB / etc

C3. Solve Real OpenCV Issue

Find issue from https://github.com/opencv/opencv/issues.

Fix it and provide test.

C4. Any Other Useful Contribution

Minimum Requirement for Level C

1. Useful technical contribution.

2. Clear benchmark / testing.

3. Professional code quality.
4. Ready for PR review.

Report Requirements

Include:

1. Which level you selected.
2. What files you modified.
3. Before / After comparison.
4. How you tested it.
5. Difficulties you met.
6. AI tools used.
7. What you learned.
8. Don't forget to include your GitHub repository link.

Rules

1. **Deadline**: Please submit your project report before its deadline. The deadline is **23:59, May 31**. After the deadline (even 1 second), **0 score!** No deadline extension for any students, even for those they enroll in the course late.
2. **Format**: Please submit the report as: *.pdf. Only one file is needed to submit. PDF format can make the layout of the report to be consistent on any computer.